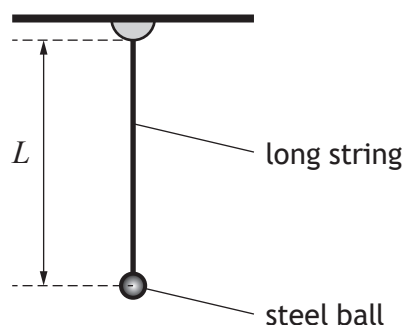


14. A student carries out an investigation to determine the gravitational field strength on Earth, using a simple pendulum.



A long string has a steel ball attached to the end of it. The length L of the pendulum can be adjusted.

The ball is raised through a small angle and then released.

The student records the time for ten complete swings and uses this to determine a value for the period T of the pendulum. The student then determines the value of T^2 .

The student repeats the experiment for different lengths.

The results are shown in the table.

L (m)	T^2 (s ²)
0.20	0.85
0.40	1.60
0.60	2.50
0.80	3.40
1.10	4.55

The gravitational field strength g can be determined using

$$\frac{T^2}{L} = \frac{4\pi^2}{g}$$

- (a) Using the square-ruled paper on *page 46*, draw a graph of T^2 against L .
(The table of results is also shown on *page 47*, opposite the square-ruled paper.)

3



14. (continued)

(b) Calculate the gradient of your graph.

2

Space for working and answer

(c) Using the gradient of your graph, determine the gravitational field strength g .

2

[END OF QUESTION PAPER]



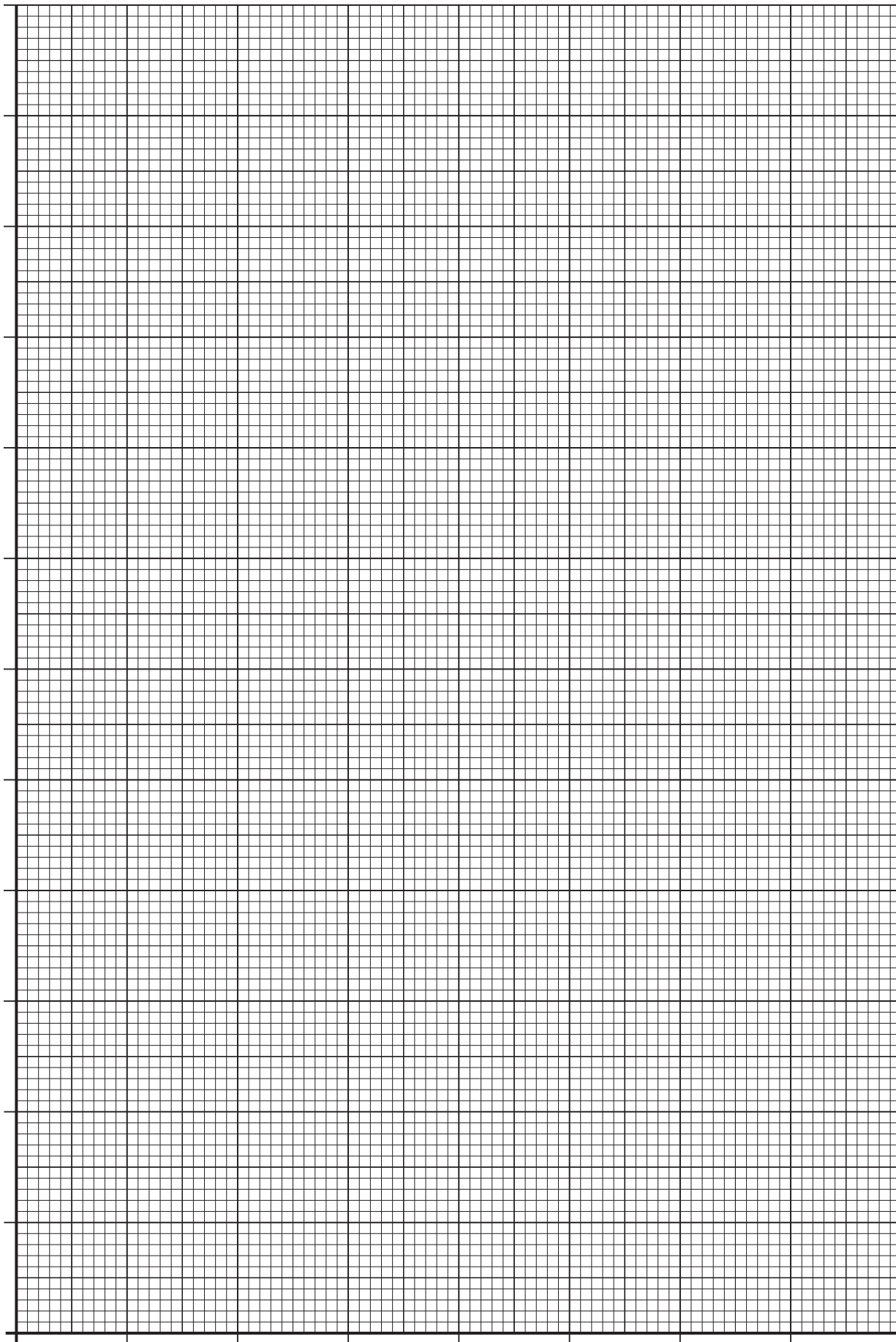
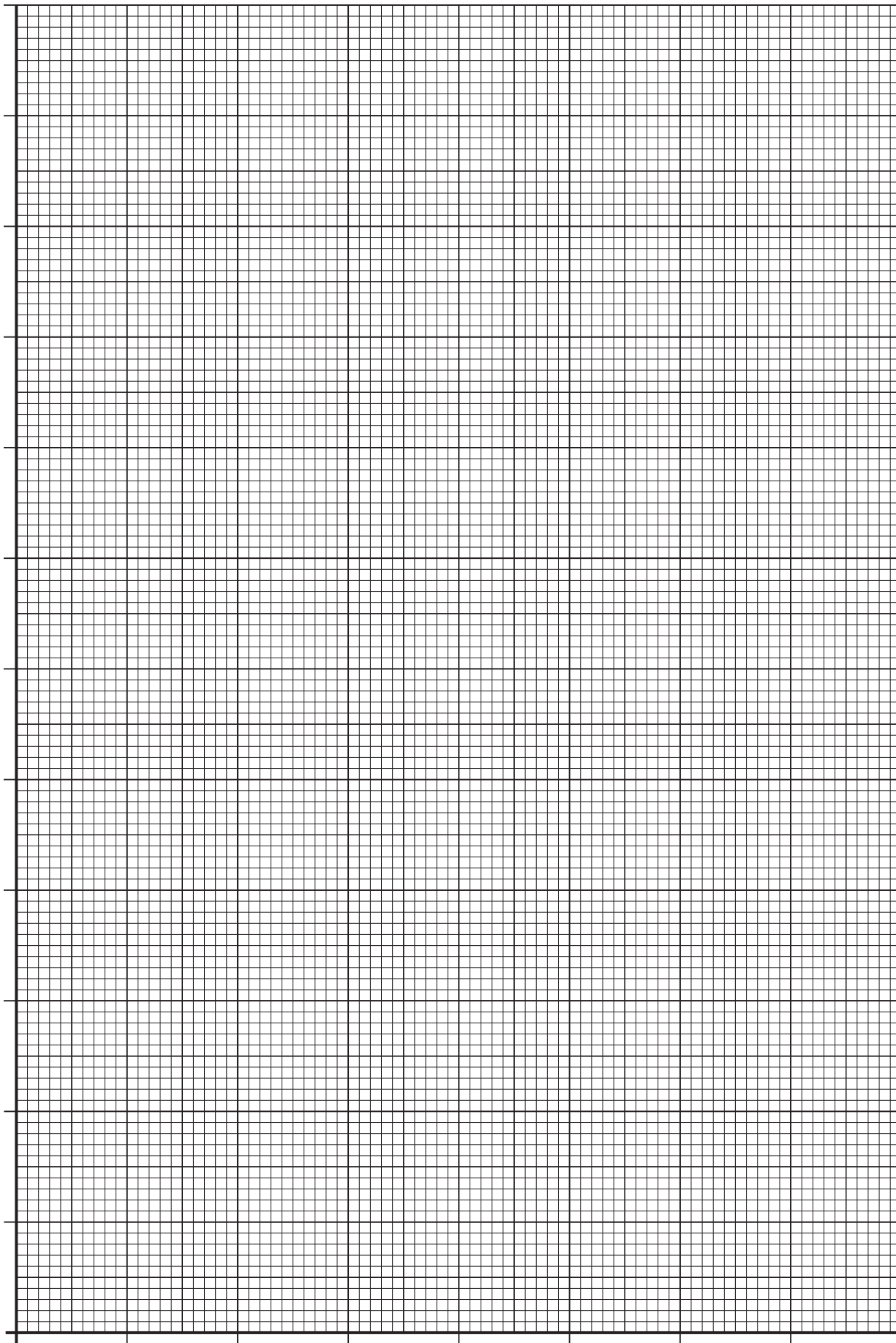


Table of results for use with question 14 (a)

L (m)	T^2 (s ²)
0.20	0.85
0.40	1.60
0.60	2.50
0.80	3.40
1.10	4.55



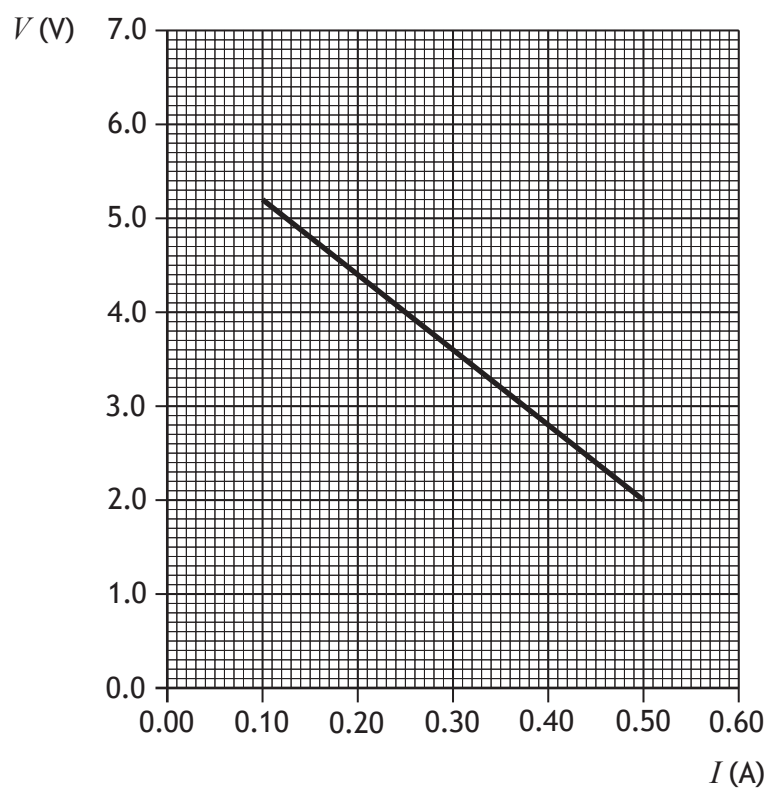
* X 8 5 7 7 6 0 1 4 7 *



* X 8 5 7 7 6 0 1 4 8 *

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional diagram for use with question 12 (e)



MARKS

DO NOT
WRITE IN
THIS
MARGIN

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK



* X 8 5 7 7 6 0 1 5 0 *

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

